Magazine Subscription Decline Analysis:

A Predictive Modeling Approach Using Logistic Regression and SVM

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Introduction

The media and publishing industry has faced sustained pressure as digital alternatives compete with traditional print subscriptions. Despite expectations that increased time spent at home would drive renewed interest in magazine readership, the company experienced a decline in subscriptions during the most recent campaign cycle. This report analyzes a marketing campaign dataset containing 2,240 customer records and 29 variables, capturing customer demographics, spending behavior across six product categories, purchase frequency across four channels, web engagement, recency of last purchase, and responses to five prior campaigns. The target variable, Response, indicates whether a customer accepted the most recent subscription offer.

Two supervised classification models are developed and compared in this analysis. Logistic regression is applied as an interpretable baseline model, offering direct insight into which variables drive subscription behavior. A Support Vector Machine model is then developed as a more flexible alternative capable of capturing non-linear patterns in the data. Both models are evaluated on accuracy, precision, recall, and AUC, and their results are compared to providing a data-driven recommendation for the company's future subscription targeting strategy.

Part 1: Data Cleansing

Dataset Overview

The dataset used in this analysis is a marketing campaign dataset containing 2,240 customer records and 29 variables. Variables include customer demographics such as year of birth, education level, marital status, and household income, as well as behavioral variables such as spending across six product categories, purchase frequency across four channels (web, catalog, store, and deals), number of web visits per month, recency of last purchase, responses to five prior marketing campaigns, and complaint history. The target variable is Response, a binary indicator (1 = subscribed, 0 = did not subscribe) representing whether the customer accepted the most recent campaign offer.

Missing Values

An inspection of the dataset revealed that only one column, Income, contained missing values, with 24 records (approximately 1.07% of the dataset) affected. Given the small proportion of missing data, median imputation was applied. The median was selected over the mean because income distributions are typically right-skewed, and the median is more robust to the influence of outliers. No other columns contained missing values.

Duplicate Records

A check for duplicate rows returned zero duplicates, confirming that every record in the dataset is unique. No rows were removed on this basis.

Irrelevant Columns

Four columns were removed prior to modeling. ID is a unique identifier with no predictive value, Dt_Customer would require additional feature engineering beyond the scope of this analysis, and Z_CostContact and Z_Revenue contained constant values across all records, providing zero variance. After removal, the dataset retained 25 columns.

Categorical Encoding

Two categorical variables required encoding before modeling. Education was ordinally encoded from 0 (Basic) to 4 (PhD), reflecting its natural rank ordering. For Marital_Status, three low-frequency categories, Alone (3 records), Absurd (2 records), and YOLO (2 records), were merged into Single as they represent individuals living without a partner. One-hot encoding was then applied using pd.get_dummies with drop_first=True, producing binary indicator variables for each remaining category. One-hot encoding was selected because marital status carries no inherent rank ordering.

Class Imbalance

The target variable revealed a significant class imbalance, with 1,906 customers (85.1%) not subscribing and only 334 (14.9%) subscribing. A naive model predicting no subscription for every record would achieve 85% accuracy without learning any meaningful patterns. To address this, class_weight='balanced' was applied in both models to ensure proportional attention to the minority class, and stratify=y was used during the train/test split to preserve the original class ratio in both subsets.

Outlier Treatment

Boxplot visualization and descriptive statistics revealed outliers in several numeric columns, particularly in spending and income variables. IQR capping (Winsorization) was applied to all numeric columns, bounding values below Q1 minus 1.5 times the IQR and above Q3 plus 1.5 times the IQR. This approach was preferred over row deletion as it preserves the full sample size while limiting the distorting effect of extreme values. A secondary boxplot review confirmed no extreme outliers remained.

Feature Scaling

Both logistic regression and SVM are sensitive to differences in feature scale. StandardScaler was applied to all features prior to modeling, transforming each variable to have a mean of zero and a standard deviation of one. This ensures that no single variable dominates model training simply due to having a larger numeric range. Scaling was fit on the training set only and then applied to the test set to prevent data leakage.

Part 2: Logistic Regression Model

Model Configuration

A logistic regression model was trained on 80% of the dataset (1,792 records) and evaluated on the remaining 20% (448 records). The model was configured with max_iter=1000 to ensure convergence, class_weight='balanced' to account for the class imbalance identified during data cleansing, and random_state=42 for reproducibility. The stratify=y option was used during the train/test split to maintain the original 85/15 class ratio in both partitions.

Model Performance

The logistic regression model achieved an overall accuracy of 76.6% on the test set. For the non-subscriber class (Class 0), the model achieved a precision of 0.96 and a recall of 0.76. For the subscriber class (Class 1), the model achieved a precision of 0.37 and a recall of 0.82. The AUC score was 0.874, indicating strong overall discriminative ability.

The high recall for Class 1 (0.82) indicates that the model successfully identifies the majority of actual subscribers. However, the lower precision of 0.37 means that among all customers the model predicts will subscribe, only 37% actually do. This is a common trade-off when using class balancing: the model casts a wider net to avoid missing subscribers, at the cost of flagging some non-subscribers as likely subscribers.

Significant Variables and Business Interpretation

The logistic regression coefficients reveal which variables most strongly influence subscription probability. The most influential positive predictor was NumWebVisitsMonth (coefficient: 0.87), indicating that customers who visit the magazine's website more frequently are significantly more likely to subscribe. This finding suggests that the company's digital presence is a key engagement channel, and that web traffic may be an early indicator of subscription intent.

NumCatalogPurchases (0.62) was the second strongest positive predictor. Customers who regularly purchase through catalogs have demonstrated a preference for curated content delivery, a behavioral trait closely aligned with magazine subscription. MntWines (0.46) and Income (0.41) both showed positive relationships with subscription, suggesting that higher-income lifestyle customers who engage in premium spending are more likely to subscribe.

On the negative side, NumStorePurchases (-0.80) was the strongest negative predictor. Customers who prefer in-store shopping may prioritize immediate, physical retail experiences over subscription-based content. Recency (-0.75) showed that customers who have not made a recent purchase are less likely to subscribe, reinforcing the idea that active engagement with the brand is a strong prerequisite for conversion. Teenhome (-0.72) suggests that customers with teenagers in the household are less likely to subscribe, possibly due to competing time and budget demands. Customers in partnered relationships (Marital_Status_Together: -0.50, Marital_Status_Married: -0.47) were less likely to subscribe compared to single customers, which may reflect different household media consumption patterns.

Part 3: SVM Model

Model Configuration

A Support Vector Machine classifier was trained using the same 80/20 train/test split and class_weight='balanced' configuration as the logistic regression model. The RBF (Radial Basis Function) kernel was selected as it is well-suited for non-linearly separable data, which is typical in customer behavior datasets. The probability=True parameter was included to enable ROC curve generation and AUC scoring. Default hyperparameters were used (C=1.0, gamma='scale'), which produced strong results without requiring grid search tuning.

Model Performance

The SVM model outperformed the logistic regression model across most metrics. It achieved an overall accuracy of 80.6%, compared to 76.6% for logistic regression. For the subscriber class (Class 1), the SVM achieved a precision of 0.42 and a recall of 0.75. The AUC score was 0.876, marginally higher than the logistic regression AUC of 0.874.

The SVM's higher precision of 0.42 versus 0.37 means it produces fewer false positives, making its subscriber predictions more reliable. The trade-off is a slightly lower recall of 0.75 compared to 0.82 for logistic regression, meaning the SVM misses slightly more actual subscribers. This reflects the SVM's tighter decision boundary, which reduces false positives at the cost of some true positives.

Feature Importance and Business Interpretation

Because the SVM uses an RBF kernel, it does not produce direct model coefficients in the same way as logistic regression. Permutation importance was therefore used to assess variable significance. This method measures the reduction in AUC when each feature's values are randomly shuffled, thereby breaking its relationship with the target variable. A larger drop in AUC indicates a more important feature.

Recency emerged as the most important variable (importance: 0.091), meaning that how recently a customer made a purchase is the single strongest predictor of subscription in the SVM model. This aligns with the logistic regression finding that disengaged customers are unlikely to subscribe. NumCatalogPurchases (0.078) was the second most important variable, reinforcing the finding that catalog channel buyers are the most subscription-prone customer segment.

NumWebVisitsMonth (0.047) and NumStorePurchases (0.045) followed, consistent with the logistic regression findings. Teenhome (0.038), MntMeatProducts (0.031), Income (0.027), MntGoldProds (0.025), NumDealsPurchases (0.023), and MntWines (0.021) rounded out the top ten. The agreement between both models on these core variables provides strong evidence that they are genuine drivers of subscription behavior rather than artifacts of any single modeling approach.

Part 4: Model Comparison and Recommendation

Performance Comparison

Model Performance Comparison on Test Set (n = 448)

Metric	Logistic Regression	SVM	Better Model
Accuracy	0.7656	0.8058	SVM
AUC	0.8740	0.8762	SVM
Precision (Class 1)	0.3716	0.4167	SVM
Recall (Class 1)	0.8209	0.7463	Logistic Regression
F1 Score (Class 1)	0.5116	0.5348	SVM

Analysis of Metrics

Accuracy measures the overall proportion of correct predictions. The SVM model achieved 80.6% accuracy compared to 76.6% for logistic regression, a meaningful improvement of four percentage points. However, given the class imbalance in the dataset (85% non-subscribers), accuracy alone is insufficient for model evaluation.

Precision for the subscriber class measures what proportion of customers predicted to subscribe actually do. The SVM achieved 41.7% precision compared to 37.2% for logistic regression. While both values appear low in isolation, this is expected in imbalanced datasets where the minority class is inherently harder to predict with high precision. The SVM's higher precision means fewer wasted marketing resources on customers who will not subscribe.

Recall for the subscriber class measures what proportion of actual subscribers the model successfully identifies. Logistic regression outperforms the SVM on this metric, achieving 82.1% recall compared to 74.6% for SVM. This means the logistic regression model catches more actual subscribers but at the cost of also flagging more non-subscribers as potential subscribers.

The F1 score balances precision and recall into a single metric, making it particularly useful for imbalanced classification problems. The SVM achieved a slightly higher F1 score of 0.535 compared to 0.512 for logistic regression, indicating a better overall balance between the two competing objectives.

Significant Variables Across Both Models

A key finding of this analysis is the strong agreement between the two models on the most important predictors. Both models identified Recency, NumCatalogPurchases, NumWebVisitsMonth, NumStorePurchases, Teenhome, Income, and MntWines as among the most influential variables. This convergence across two fundamentally different modeling approaches provides high confidence that these variables are genuine drivers of subscription behavior.

The consistency of these findings carries important business implications. The magazine company should prioritize outreach to customers who have purchased recently, visit the website frequently, and purchase through catalogs. These customers represent the highest-value subscription targets. Conversely, customers with teenagers at home, those who primarily shop in-store, and those who have not engaged with the brand recently represent lower-probability targets and may warrant a different marketing strategy or reduced investment.

Model Recommendation

Based on the three primary evaluation metrics, the SVM model is recommended as the primary predictive model. It achieves higher accuracy (80.6% vs 76.6%), higher precision (41.7% vs 37.2%), and a higher F1 score (0.535 vs 0.512), making it more reliable for identifying true subscribers while minimizing wasted outreach.

The one area where logistic regression outperforms SVM is recall (82.1% vs 74.6%). If the company's strategic priority is to capture every possible subscriber regardless of the cost of false positives, logistic regression would be the preferred choice. This would be appropriate in a scenario where the cost of missing a potential subscriber is very high, such as during a low-cost digital campaign where broad targeting carries minimal marginal cost.

However, for campaigns involving direct mail, catalog distribution, or other high-cost outreach, the SVM's superior precision makes it the more economically efficient choice. Additionally, logistic regression remains a valuable complementary tool for communicating findings to business stakeholders who need to understand the directional effect of specific variables, something that SVM's permutation importance cannot provide with the same clarity.

In summary, the magazine company's subscription decline can be addressed by targeting high-income, recently active, web-engaged, catalog-purchasing customers, particularly those without teenagers in the household. Both models agree on this profile, and the SVM model provides the most accurate and precise means of identifying these customers at scale.

References

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Appendix

The following figures were generated during the analysis. All visualizations were produced using Python with the matplotlib and seaborn libraries.

Figure 1. Class distribution of the Response variable (85% non-subscriber, 15% subscriber).

```
Response
0    1906
1     334
Name: count, dtype: int64
Response
0    0.85
1    0.15
Name: proportion, dtype: float64
```

Figure 2. Boxplots of all numeric features following IQR-based outlier capping.

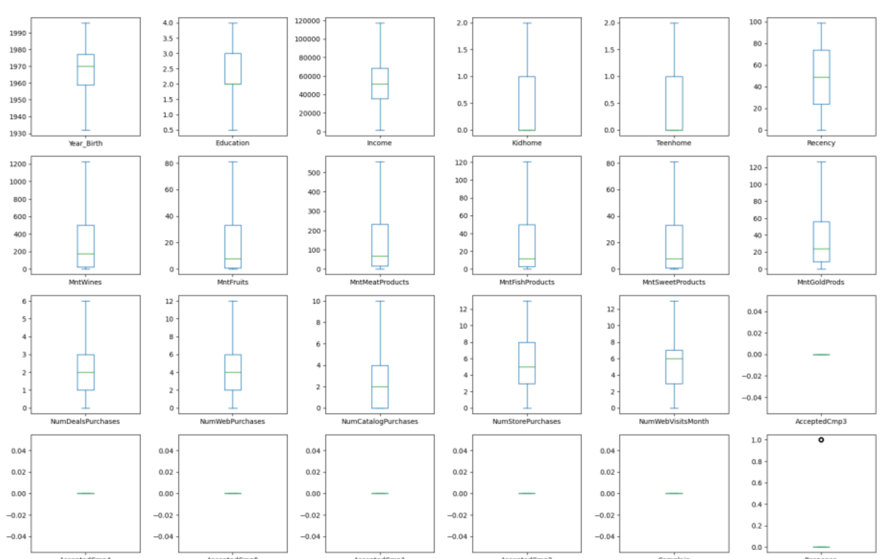

Figure 3. Logistic Regression: Confusion matrix and ROC curve.

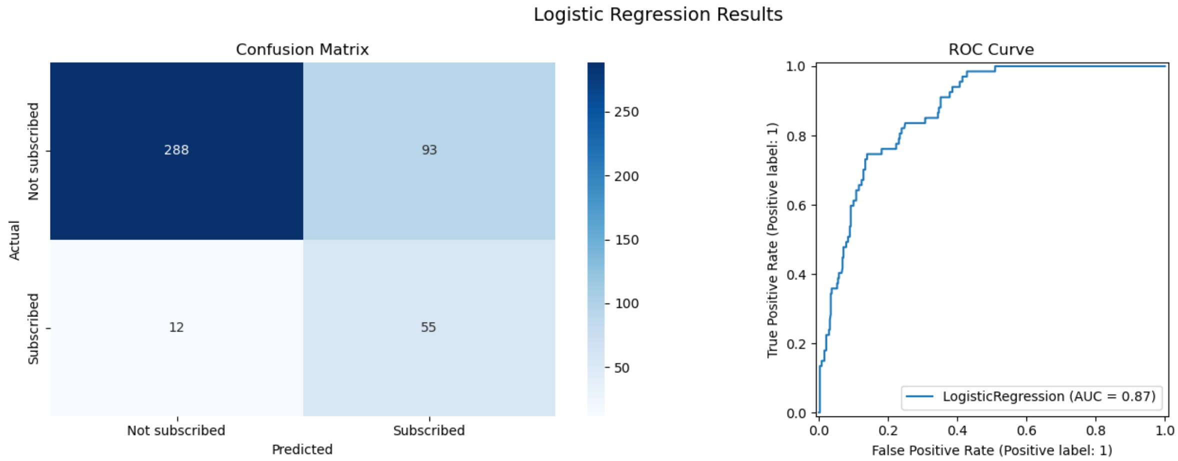

Figure 4. Logistic Regression: Top 10 feature coefficients (horizontal bar chart).

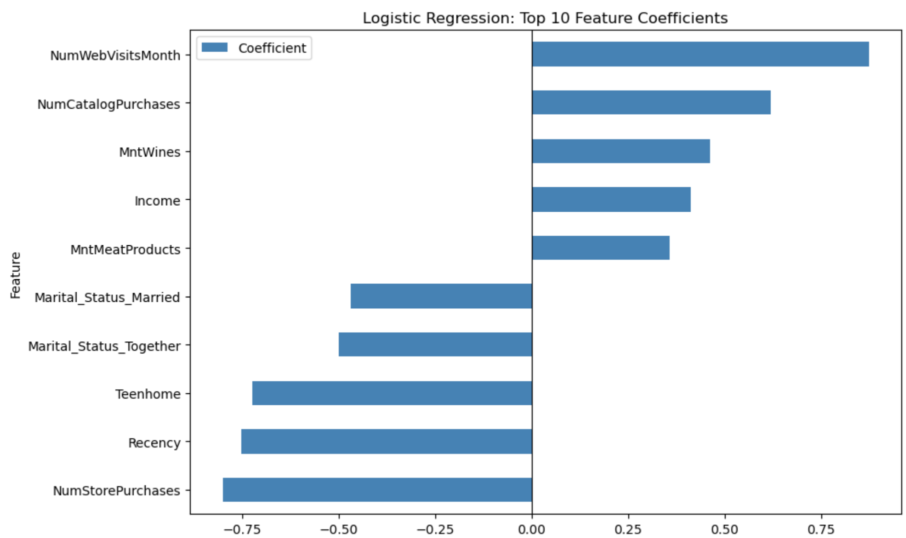

Figure 5. SVM: Confusion matrix and ROC curve.

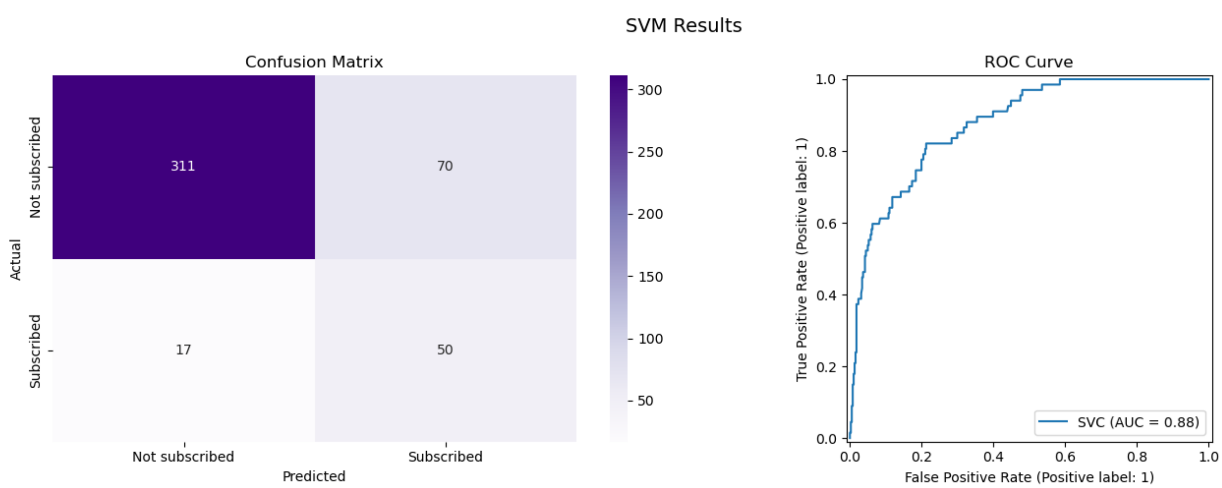

Figure 6. SVM: Top 10 permutation feature importance scores (horizontal bar chart).

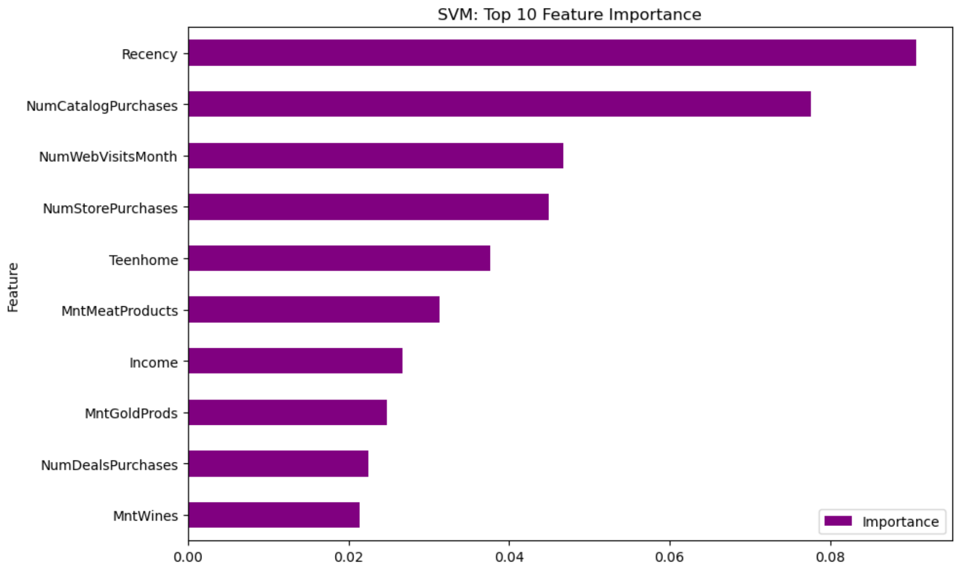

Figure 7. Model comparison bar chart: Accuracy, AUC, Precision, Recall, and F1 Score for both models.